

Statistical modeling of large-scale public Wi-Fi hotspots

Poisson regression modeling and queuing analysis of:

- Arrival counts
- Connection duration
- Simultaneously present #of customers

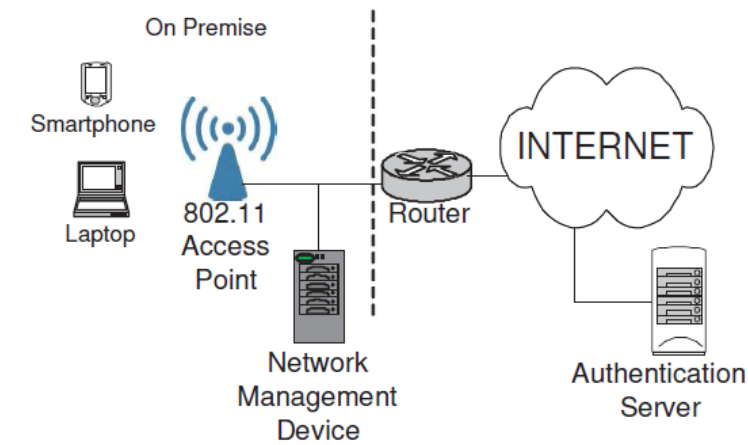
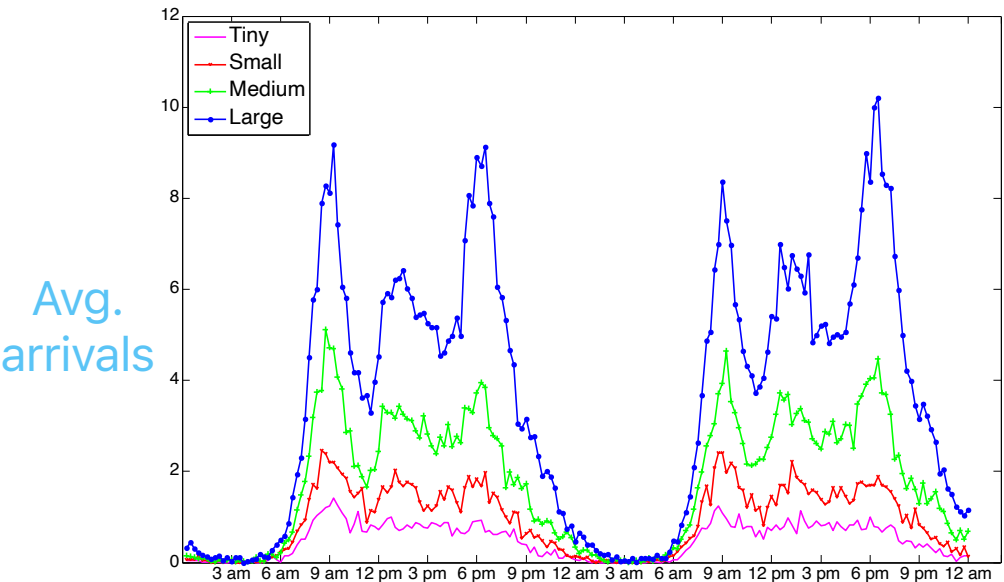


Fig 1. Mobile Internet access using Wi-Fi hotspots

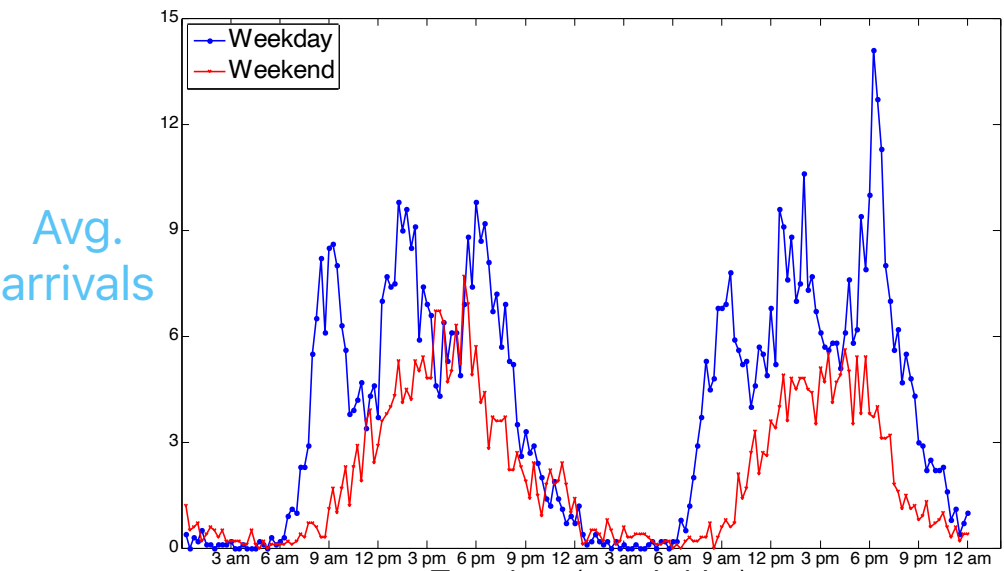
Connection Duration (min)	Mean	SD
Coffee shops, fast food chains	29.8	81.9
Book stores, hotels	73.4	142.3
Enterprises, stadiums	61.6	113.8

Fig 2. Characteristic peaks and variation in the average number of arrivals for:

(a) 60 coffee shops in each category of tiny, small, medium, and large over two weekdays



(b) 20 book stores and hotels over two weekdays and weekend



Weekdays show three peaks during morning, noon, and evening, whereas weekends show only one peak during late afternoon.